Sarah Sameera Peruka

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Summary

Computer Science undergraduate pursuing a B.Tech in Computer Science Engineering and a BS in Data Science from IIT Madras. Strong foundation in **Java, Python, Data Structures & Algorithms, OOP, DBMS, and Computer Networks**, with hands-on experience developing full-stack applications, backend services, REST APIs, authentication systems, and database-driven solutions. Experienced with **React.js, Node.js, Express.js, Flask, MongoDB, and SQLite**, with exposure to Git, Linux, Docker, testing, deployment, and AI technologies.

Education

- **B.Tech in Computer Science Engineering, BVRIT Hyderabad College of Engineering for Women**: CGPA: 8.65 2023–2027
- **BS in Data Science, Indian Institute of Technology Madras**: CGPA: 7.5 2023–Present

Technical Skills

Languages: Java, Python, JavaScript, C, SQL

Core Programming: Data Structures & Algorithms, OOP, Problem Solving, DBMS, Operating Systems, Computer Networks

Software Engineering: REST APIs, CRUD, API Integration, JWT Authentication,Debugging

Web & Backend: React.js, Node.js, Express.js, Flask, HTML5, CSS3, Bootstrap

Databases: MySQL, MongoDB, SQLite

AI/ML: LangChain, RAG, FAISS, LLM APIs, Vision-Language Models,Federated Learning

Tools & Deployment: Git, GitHub, Docker, Postman, Vercel, Render

Experience

- **Research Intern – IIT Hyderabad**: Worked on privacy-preserving BI-RADS breast density classification using mammography datasets. Fine-tuned pre-trained **Vision-Language Models (VLMs)** and investigated **Federated Learning** approaches including FedAvg for distributed model training and evaluation.

Projects

- **To-Do List Application – MERN Stack *GitHub***: Built and deployed a full-stack application using **React.js, Node.js, Express.js, and MongoDB**. Developed **REST APIs** for CRUD operations with **JWT authentication**; implemented search, filtering, priorities, categories, and due-date management with persistent database operations.
- **Placement Portal Application – IIT Madras BS Degree Project *GitHub***: Developed a **Flask-based web application** using **Python, Jinja2, SQLite, HTML, and Bootstrap** to automate placement workflows. Implemented **role-based access, database operations, workflow management, search, approvals, resume handling, job applications, and dashboards** for students, recruiters, and administrators.
- **Sentinel Secure AI – Financial Fraud Analysis Platform *GitHub***: Developed a modular **AI-powered financial security application** using **Python, Streamlit, SQLite, and LangChain**. Implemented **authentication, transaction management, BIN-based card analysis, fraud risk scoring, and document Q&A** using modular authentication, database, fraud-analysis, and RAG components.
- **Tranquilink – Smart India Hackathon Round 1 Qualifier *GitHub***: **Team Leader** for a hackathon project; coordinated task allocation and feature integration while contributing to frontend development and application workflows. Built a platform with **mood tracking, anonymous peer interaction, personalized resources, and counselor workflows** under time constraints.
- **Healthcare Analytics Dashboard *GitHub***: Built a **Python, SQL, and SQLite** workflow involving data processing, relational database creation, SQL querying, exploratory analysis, and Power BI visualization.

Certifications & Achievements

Certifications: Microsoft Azure DevOps Boards – Coursera; Full Stack Web Development – Udemy; HackerRank – SQL (Intermediate); HackerRank – Python; IIT Madras BS Degree Foundation Certificate

Hackathons & Leadership: Smart India Hackathon- Round 1 Qualifier(Team Leader); Flipkart GRID 8.0 – Advanced to Round 3; Adobe India Hackathon – Round 1 Qualifier

Problem Solving: Strong foundation in **Data Structures & Algorithms, Java, Python, OOP, DBMS**, demonstrated through competitive coding(430+ LeetCode problems solved), academic projects, and hackathon development.

Additional Information

Languages: English, Telugu, Hindi, French

Extracurricular: Black Belt in Karate; National Gold Medalist – KATA; CSI Student Chapter Member; NSS Volunteer